

# Disproof of Conjecture 00000004410

## Equal sampling laws force equal entropy rates

gaochengzhecpu

October 4, 2026

### Abstract

The conjecture proposes two graphons with identical sampling distributions at every finite window size but different entropy rates. Under the given definition of entropy rate, this is impossible: the finite entropies and their normalizations agree term by term, so their limits agree. The argument works for arbitrary finite sampling spaces and every common positive normalization, without a graphon reconstruction theorem.

## Definitions and proof

For a window count  $n$ , let  $\Omega_n$  be the finite set of possible sampled outcomes and let  $p_n^W$  be the probability mass function generated by a graphon  $W$ . For ordinary labeled graph samples,  $\Omega_n$  is the set of simple graphs on  $n$  labeled vertices. More generally, the following proof uses any common finite outcome set for each window count.

The Shannon entropy of the finite sampling law is

$$H_n(W) = - \sum_{\omega \in \Omega_n} p_n^W(\omega) \log p_n^W(\omega), \quad 0 \log 0 := 0.$$

Write  $c_n > 0$  for the shared normalization used to define the entropy rate:

$$h(W) = \lim_{n \rightarrow \infty} \frac{H_n(W)}{c_n},$$

when this real limit exists. The per-window convention in the source uses  $c_n = n$  for  $n \geq 1$ . The proof also applies to other shared positive normalizations; changing a finite initial segment or reindexing positive window counts does not change the rate.

**Proposition 1.** *If  $p_n^U = p_n^V$  for every finite window count  $n$ , and both entropy rates exist under the same convention, then  $h(U) = h(V)$ .*

*Proof.* For each  $n$  and each outcome  $\omega$ , the probabilities agree. Substitution in the finite sum gives  $H_n(U) = H_n(V)$ . Division by the same  $c_n$  yields

$$\frac{H_n(U)}{c_n} = \frac{H_n(V)}{c_n} \quad \text{for every } n.$$

The two entropy-rate sequences are therefore the same sequence. Uniqueness of limits in  $\mathbb{R}$  gives  $h(U) = h(V)$ .  $\square$

The conjecture requires unequal rates, indeed a strictly positive constant gap. But the proposition forces their difference to be 0. Consequently the required pair of graphons cannot exist.

## Why this reduction is sufficient

The proof uses equality of the *entire finite sampling laws*, as stated in both language versions. Equality of selected moments, selected graph statistics, or a single fixed window size would be a different hypothesis. Likewise an entropy of unobserved latent variables would be a different quantity from the sampling entropy rate defined in the source.

No injectivity or reconstruction result for graphons is required. Assigning a graphon its finite sampling laws maps the proposed pair into the class of sequences of finite probability distributions. The proposition rules out separation for *every* such sequence, a larger class than those realized by graphons. There is no need to assume or prove that every sequence is graphon-realizable. This universal impossibility, not a finite enumeration or a sampling experiment, is the disproof.

## Lean verification

The accompanying project defines real-valued probability masses with proved nonnegativity and total mass one, and Shannon entropy using the real logarithm. The theorem for arbitrary finite outcome spaces proves equality of normalized entropy-rate limits directly. A graph-specific instantiation encodes a labeled simple graph by one Boolean variable per unordered pair of distinct vertices. The main negation rules out two such sampling-law sequences with the stated equal-law and unequal-rate properties. A further theorem allows any type of objects mapping to those laws, including graphons.

The file also constructs deterministic empty-graph laws and proves their entropy rate is zero, so the probability-law and limit definitions have an explicit example. It does not construct a separate analytic graphon library; the universal finite-law theorem is already sufficient for the reduction above. The Python script only checks finite examples and is not a premise of the proof.

Lean 4.19.0 and Mathlib v4.19.0 are pinned, with Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The complete project, reproduction commands, actual build logs, printed foundational axiom dependencies, source and PDF hashes, and adversarial self-review accompany this submission. The exact bilingual conjecture is preserved in `SOURCE.md`.